# Explaining the better move

17 rounds of hill-climbing a game review's lessons, graded on the board and by a blind judge

Chess Coach research lab

**Abstract**

At each key moment of a game review, the coaching app explains what the move played allowed, what the engine's better move does instead, and the difference between them. We measured those explanations on the 329 key moments of 60 of the player's games, and an endgame round also on 403 endgame moments of 160 more, with three grades checked against the board, and changed the code that writes them one idea per round, keeping a round only when no grade fell on either half of the games (one round fixed five text defects at once). Where a change wrote something the grades cannot check, a judge model compared the old and new texts blind, twice, with the order reversed. Over 17 rounds (14 kept), the share of held-out moments whose texts make no false checkable claim rose from 90.1% to 100.0%, and the share that names a concrete consequence of the better move from 75.0% to 82.9%. The judge preferred the texts of every kept round it read in most of the comparisons it decided on each half, and a length control showed that was not for their length: filler as long as a round's change won 0 of 234 comparisons against the same text without it. Each kept round is a change to the app's own code. The lab recomputes every published number from the records, and the app rerun on fresh engine evidence, carrying the kept rounds through round 16, scores 100.0% and 84.3% on the same two measures.

## 1 What the player reads

When a game review stops at a key moment (a mistake, a blunder, a missed win), the app shows a card with a headline and a verdict, a lesson panel (what the move played allowed, what the better move does instead, the difference between the two lines, a takeaway), a Why sheet with the evidence behind the verdict, and a retry: the player may try the position again, with a hint and then the answer. Every one of those texts is written by heuristics in the app's review and knowledge services from the engine's evidence. No language model writes them, so every claim they make can be traced to a template, and most can be checked on the board.

The player is an adult club player, rated 948–1167 on chess.com across the 60 games studied here. Two things matter: that the texts are true, and that they say *why* the better move is better, in terms the player can use at the board, not only by how much.

## 2 Data and measures

**Moments.** The 329 key moments the review stops at in 60 of the player's games, from the app's phone-profile engine evidence for each game. The games are split 30 for train and 30 for test, the same split as the accuracy project, so no game is train in one and test in the other: 177 train moments and 152 test moments. A round was designed on train moments only.

**Grades.** Each moment's texts get three grades, all checked against the board with chess.js and against the stored engine lines (`grade.mjs`):

- **faithful**: 1 unless a checkable claim in the visible text is false: an illegal move, a line that does not replay, a piece said to attack what it does not, material said to be won that the line does not win, a mate that is not there, or an "if they take it with" answer that is not the engine's line or whose claim does not hold at its end.
- **concrete**: 1 when the better move's explanation names a consequence (material, a mate, a named threat, a prevented reply) and the text says what the move played allowed or missed.
- **named_deep**: 1 when every move the text names as best rests on a search at the app's best budget. It is a property of the evidence, not of the words, and is held, not targeted.

A fourth grade, **faithfulAll**, is faithful over the lesson the Why sheet also shows on a moment whose card shows only a comment. A clause no template recognises is reported as unparsed, never graded true. The grader was checked by hand on 19 moments before any round; the hand verdicts and the baseline's grades agree on all three grades for 19 of them.

Every rate below is a share of key moments, with a 95% interval from a bootstrap over games (a game's moments share its evidence, so the game is the unit). A variant's difference from the baseline is paired on the same moments.

**The judge.** The grades cannot see whether a new clause *helps*. For a round that writes something new, every moment whose texts change is shown to a judge model as two versions (X and Y, by a seeded coin), each laid out as the app lays it out, with the position, the move, the verdict, the engine's evaluation and best line. The judge answers which version would help the player more (understand why the better move is better, and know what to look for next time), or that they are the same, with one sentence of reason. It answers twice, with X and Y swapped the second time. The rule, fixed before any run: keep the round if, pooling both runs, the new texts are preferred in more than half of the decided comparisons on train *and* on test.

## 3 The rounds

Table 1 lists every variant. Rounds 1 and 2 fix false claims the baseline's templates made, found by the grader over all moments before the split was used to design anything. Rounds 3 to 10 add something the player asked for on the app's Coach Calibration page or that the judge's notes named, so the judge decides them. Round 11 fixes five defects in round 10's texts that the player's calibration answers and the judge's notes flagged; each can be read off the text and the board, so the round counts them rather than asking the judge. Round 12 answers the question the player's second set of calibration answers kept asking, what happens when the opponent simply saves the attacked piece, so the judge decides it. Round 13 shortens round 12's sentence where the judge's notes said it ran long, so the judge decides it too. Round 14 corrects how the lessons name an amount: its number was the line's net, but the pieces it named were not always the ones that changed hands. That is read off the board, which the judge does not see, so the round counts its defects as round 11 did. Round 15 fills the sections on what the played move allowed that named the reply and said nothing about it, which the player's own calibration answers had pointed to, so the judge decides it. Round 16 tells an endgame's passed pawns' story, which the player's calibration answers asked for in his rook endings; the 60 games hold too few endgame moments to decide it, so it is also graded and judged on an endgame set of 160 more games (Section 6), and the judge decides it. Round 17

says where each move leaves the game, which the player's third set of calibration answers asked about; it changes 347 moments, more than a judge can read, so the judge reads a seeded sample of them and decides it.

| | | test (%) | | train (%) | | texts | judge |
|---|---|---|---|---|---|---|---|
| variant | change | faithful | concrete | faithful | concrete | changed | train / test |
| baseline | the app before the rounds | 90.1 | 75.0 | 87.6 | 78.0 | | |
| v1 | Say each line's material as it is in the lesson's difference | 96.1 | 82.2 | 98.3 | 84.7 | 20, 34 | |
| v2 | Name only the pawns the seventh-rank rook hits | 100.0 | 82.2 | 99.4 | 84.7 | 25, 35 | |
| v3 | Say what the better move defends | 100.0 | 82.2 | 99.4 | 84.7 | 31, 43 | 22–0 / 18–2 |
| v4 | Answer the capture a learner asks about first | 100.0 | 82.9 | 99.4 | 84.7 | 34, 45 | 8–0 / 6–0 |
| v5 | Say where a long mate ends (*not kept*) | 100.0 | 82.9 | 99.4 | 84.7 | 45, 54 | 6–6 / 14–2 |
| v6 | Say where a long mate ends, when the end is in reach | 100.0 | 82.9 | 99.4 | 84.7 | 44, 51 | 4–0 / 10–0 |
| v7 | Say what a capture the better line declines would cost (*replaced by v7b*) | 98.7 | 82.9 | 97.2 | 84.7 | 65, 77 | 48–3 / 40–1 |
| v7b | Say what a capture wins, not the balance it leaves | 100.0 | 82.9 | 99.4 | 84.7 | 65, 77 | 51–0 / 41–0 |
| v8 | Say when the played move is right but too early | 100.0 | 82.9 | 99.4 | 84.7 | 70, 87 | 50–0 / 20–0 |
| v9 | Answer the natural capture, and skip what a pawn plainly guards (*not kept*) | 100.0 | 82.2 | 98.9 | 85.9 | 142, 166 | 1–35 / 0–12 |
| v10 | Say the natural capture that only trades as the trade it is | *graded on rebuilt documents (Table 4)* | | | | | 18–2 / 12–0 |
| v11 | Five text defects, fixed where the texts are written | *graded on rebuilt documents (Table 4)* | | | | | |
| v12 | Say what the natural defense to the better move costs | *graded on rebuilt documents (Table 4)* | | | | | 19–9 / 22–12 |
| v13 | Leave out the moves the defense sentence does not need (*not kept*) | *graded on rebuilt documents (Table 4)* | | | | | 0–16 / 4–24 |
| v14 | Say an amount in the pieces it is made of | *graded on rebuilt documents (Table 4)* | | | | | |
| v15 | Say what a quiet reply defends | *graded on rebuilt documents (Table 4)* | | | | | 22–1 / 8–0 |
| v16 | Tell an endgame's passed pawns' story | *graded on rebuilt documents (Table 4)* | | | | | 37–2 / 18–3 |
| v17b | Say where each move leaves the game | *graded on rebuilt documents (Table 4)* | | | | | 73–0 / 76–0 |

Table 1: Every variant. Rates are shares of key moments; *texts changed* counts the test and train moments whose visible texts differ from the baseline's; *judge* is the round's pooled preferences, new–old, on train and test (Table 2).

**Round 1: each line's material as it is.**

The lesson's fallback sentence said "neither line wins material in the next 5 moves" whenever no earlier branch fired, including when both lines won the same material. It now says each line's material.

**Round 2: the seventh-rank rook.**

The insight named pawns the rook could not see past another pawn. It now names only the pawns the rook hits.

**Round 3: what the better move defends.**

A piece the reply takes, guarded or moved away by the better move (Figure 1).

**Round 4: the capture a learner asks about first.**

Where the better move or the reply lands on a square it can be taken on and the engine's line takes it, the lesson says what follows (Figure 2).

**Rounds 5 and 6: where a long mate ends.**

A mate is told with where it ends and what holds it there. Round 5 did so for every mate of three moves or more and failed its rule on train; round 6 keeps it for mates of three to seven moves with no promotion on the way.

**Round 7: a capture the better line declines.**

A search of each capture the line declines (`probes.mjs`, at the routine's budget), and the lesson says what it would cost: a recapture as the trade it is, or checks that win a piece, move by move (Figure 3). Round 7b fixes the material clause the judge found false in a few moments (a gain read as a balance).

**Round 8: the right move, too early.**

Where the move played is in the engine's best line at the mover's next or second turn, the difference and the takeaway say so, with the line (Figure 4).

**Round 9 (not kept): fewer capture answers.**

Skipping an answer when a pawn plainly guards the square, and stopping at the first capture with no answer. The judge wanted every answer back (Figure 5).

**Round 10: the natural capture that only trades.**

The cheapest capture of the better move's piece is said as the trade it is when its search takes back at once on the same square, nothing takes there again, and the material ends where it began: "if they take it with axb5, axb5 takes back: an even trade" (Figure 6). The round was graded on documents rebuilt for it (Section 6).

**Round 11: five text defects.**

Five errors in round 10's texts, fixed together because each is a mistake to correct, not an idea to test: the retry's answer opened its later sentences in lower case; a reply that only takes back what the move captured was called a win "outright"; a capture or a check was called "a quiet improvement"; the difference called the lines "too short or unsettled" to compare beside a capture the same card calls outright; and a line's net gain was glossed with a piece worth more than it, as in "they win a piece and a pawn (your rook on d7)". The round was graded on round 10's rebuilt documents (Section 6).

**Round 12: what the natural defense costs.**

When the better move threatens a piece, a human-move model gives the defender's likeliest reply at the defender's rating band. When that reply moves the attacked piece or blocks the attack, the engine's line from it (the better line itself when it is the engine's reply there, otherwise a search at the routine's budget, `defenses.mjs`) says what it costs: "if they move the rook (Rf7), Qxf5, and you have won a pawn" (Figure 7). A capture of the attacker is left to the capture answers, and the sentence appears only when the line wins material after the defense or mates. The round was graded on documents rebuilt for it (Section 6).

**Round 13 (not kept): the defense sentence, shorter.**

Where the defense is the engine's own reply, the sentence points to the engine's line instead of printing its moves again ("if they move the bishop (Bb2), as in the engine's line, you have won a pawn by move 14"); where the line would show more than 4 moves, it says only when

the gain is in ("if they move the queen (Qxd5), you have won about 6 pawns of material by move 19"). The judge wanted the moves back (Figure 8).

**Round 14: an amount in the pieces it is made of.**

The lesson turned a line's net material into a count whatever pieces made it up, so a bishop won for two pawns was "a pawn" and a rook won for a pawn "a piece and a pawn"; the same count picked the piece said to fall, and some lines gave that piece back. Each line now carries the pieces its mover won and gave, by kind, with trades set off (a piece against one of its kind, then a knight against a bishop), and an amount keeps its count only when the count names those pieces. Otherwise it names them: "they win a bishop for two pawns", "limits the loss to a rook for a pawn". Where one sentence gives an amount for two lines whose pieces differ, it says the worth ("a pawn's worth of material"), and a piece is said to fall only when the line keeps it. The round was graded on round 13's documents (Section 6).

**Round 15: what a quiet reply defends.**

Where the section on what the played move allowed named the reply and gave no reason for it, and the reply captures nothing, the lesson says what the reply defends, checked on the board: it moves or guards the piece the better move would have won or attacked ("it moves the rook off e5, where dxe5 would have won it"; "it guards the pawn on e4, which Nxe4 would have won"), takes away the better move's check ("it takes away the check Qd3 would have given"), or answers the played move's own threat ("it guards the pawn on b5, which Rd5 attacked"). The headline's sentence about the reply carries the point. The round was graded on the documents round 14 was (Section 6).

**Round 16: an endgame's passed pawns.**

In an endgame, and when neither line ends in mate, the lesson follows each pawn over the lines it reads. The pawn that queens after the played move and not after the better one opens the section on the difference with its steps and its fate in the other line ("After Kg4, their b-pawn queens: b7 and b8=Q in the engine's line. Nxb6 takes it first."; "After Rc3 it falls (Rxh2)"); a reply that pushes that pawn "goes on to queen"; a better move that takes a passed pawn, or stands in the path of one past the middle of the board, says so ("it blocks their passed b-pawn on b7"); and a king's walk to a passed pawn, each move closer, is named with its moves ("your king walks over to their c-pawn: Kb2, Kc3 and Kxc4"). The shown lines run on to the promotion or the walk's last move. The round was graded on the documents round 15 was and on the endgame set (Section 6).

**Round 17: where each move leaves the game.**

On an inaccuracy, mistake, miss or blunder, the section on the difference ends with where the played move and the better one leave the mover, in the words and the number the app gives a score, from the mover's side. When both moves leave the mover ahead (much better or winning) or behind (much worse or losing), it says so ("You are behind either way: losing after Rb5 (−8.1), much worse after Rxb2+ (−4.4)."); on a mistake, miss or blunder whose better move leaves a better standing, it says where each leaves the mover ("After Kg4 you are losing (−7.0). After Nxb6 you would be winning (+7.0)."). It says nothing when both moves leave the game open, from worse to better. Round 17 first wrote the second form as one sentence joined by a semicolon, which the app's rules for its words forbid; round 17b writes it as two. The round was graded on the documents round 16 was, in both sets (Section 6).

| round | judged | train | test | the rule says |
|---|---|---|---|---|
| 3 | round 3 | 22–0 (0) | 18–2 (0) | keep |
| 4 | round 4 | 8–0 (0) | 6–0 (0) | keep |
| 5 | round 5 | 6–6 (10) | 14–2 (12) | do not keep |
| 6 | round 5 (17 moments) | 4–0 (8) | 10–0 (12) | keep |
| 7 | round 7 | 48–3 (15) | 40–1 (9) | keep |
| 7b | round 7b | 20–0 (0) | 12–0 (0) | keep |
| 7 and 7b | round 7 then round 7b | 51–0 (15) | 41–0 (9) | keep |
| 8 | round 8 | 50–0 (0) | 20–0 (0) | keep |
| 9 | round 9 | 1–35 (0) | 0–12 (0) | do not keep |
| 10 | round 10 | 18–2 (0) | 12–0 (0) | keep |
| 12 | round 12 | 19–9 (0) | 22–12 (0) | keep |
| 13 | round 13 | 0–16 (0) | 4–24 (0) | do not keep |
| 15 | round 15 | 22–1 (1) | 8–0 (0) | keep |
| 16 | round 16 | 37–2 (3) | 18–3 (1) | keep |
| 17 | round 17 (80 moments) | 72–0 (8) | 74–0 (6) | keep |
| 17b | round 17b (49 moments) | 52–0 (0) | 46–0 (0) | keep |
| 17 and 17b | round 17 then round 17b (80 moments) | 73–0 (7) | 76–0 (4) | keep |

Table 2: The judge rounds, pooled over both runs: new–old preferred, with the same-answers in parentheses. Round 6 made no run of its own: it reads round 5's judgments on the moments whose texts round 6 changes. Round 17's judges read a seeded sample of the moments it changed; round 17b judged again the sampled moments whose wording it changed, and round 17 is decided on both.

## 4 Is the judge rewarding length?

Before round 10, the judge picked the longer of its two versions in 305 of the 326 comparisons it decided in rounds 3, 4, 5, 7, 7b, 8 and 9. A judge that simply prefers longer text would have voted the same way, so a length check, fixed before any of its answers, tested that on the moments of rounds 7, 8 and 9. Each moment gave three comparisons: the longer text against itself; a *decoy*, the shorter text with filler about as long as the round's change, against the shorter text; and the longer text against the decoy. The filler either restates what the card already shows (its line, or that the move is the engine's first choice) or is advice that fits any move. The last two comparisons were shown in both orders, and the 585 items were dealt into 20 batches under opaque ids, each read by a fresh judge with the rounds' instructions, so no judge could tell a control from a round.

The judge answered "same" to 117 of 117 identical pairs, and preferred the decoy in 0 of 234 comparisons with the shorter text (Table 3). The rounds that added words won on what the words said. From round 10 on, each round's batches also carry a padding control: moments the round does not change, set against themselves with filler as long as the round's change. If the decoy wins at least a quarter of those comparisons and more than half of the decided ones, the judge has drifted toward length and the round is not kept on its word. In round 10 the decoy won 0 of 20, and in round 12 0 of 20. Round 13 set out to shorten its texts, so its batches carry the mirror, a trim control: moments the round does not change, set against themselves with the last point of the better move's section cut. A judge drifting toward short texts would prefer the cut one, under the same rule; it did in 1 of 20 comparisons (16 the full text, 3 same). Round 15 grows the section on what the played move allowed, so its padding control pads that section, on moments where it says nothing and the round leaves it so; the decoy won 0 of 20 comparisons (17 the shorter text, 3 same). Round 16 grows several sections of the card at once, so its padding control adds a sentence of filler after the difference, as long as the round's median growth of the whole card, on endgame

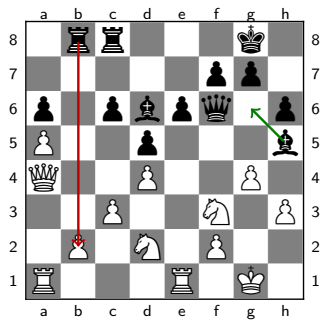
moments the round leaves unchanged; the decoy won 0 of 20 comparisons (14 the shorter text, 6 same). Round 17 adds one sentence at the end of the difference, so its padding control does as round 16's did, on moments of the endgame set the round leaves unchanged; the decoy won 0 of 20 comparisons (0 the shorter text, 20 same).

| check | result | verdict |
|---|---|---|
| J1, the same text on both sides: answered "same" | 117 of 117 | uses "same" |
| J2, the shorter text against itself with filler: the decoy preferred | 0 of 234 (232 the shorter text, 2 same) | not length-biased |
| J3, round 7: the longer text against a decoy of its length | train 66 of 66, test 50 of 50 | survives |
| J3, round 8: the longer text against a decoy of its length | train 50 of 50, test 20 of 20 | survives |
| J3, round 9: the longer text against a decoy of its length | train 36 of 36, test 12 of 12 | survives |
| padding control, round 10: the decoy preferred | 0 of 20 (20 the shorter text, 0 same) | no drift |
| padding control, round 12: the decoy preferred | 0 of 20 (20 the shorter text, 0 same) | no drift |
| trim control, round 13: the decoy preferred | 1 of 20 (16 the full text, 3 same) | no drift |
| padding control, round 15: the decoy preferred | 0 of 20 (17 the shorter text, 3 same) | no drift |
| padding control, round 16: the decoy preferred | 0 of 20 (14 the shorter text, 6 same) | no drift |
| padding control, round 17: the decoy preferred | 0 of 20 (0 the shorter text, 20 same) | no drift |

Table 3: The judge's controls, recomputed from their answers (`judge/controls/score.mjs`). In round 9 the longer text is the old one, so its surviving the decoy means round 9's rejection holds.

## 5 Examples

Each example is the first moment of its round, in the round's own order, on which both runs of the judge agreed. The tables show only the sections whose text the round changed.

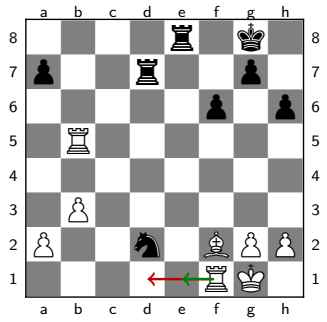


**Played:** 18. . . ♖xb2, Black to move; the app's verdict: Mistake.
**Engine:** +1.8 with the best move, +0.1 after yours.
**Best line:** 18. . . ♗g6 19. b4 h5 20. ♘e5 hxg4 21. ♘xg4
**UCI:** `b8b2`; best line `h5g6 b2b4 h6h5 f3e5 h5g4 e5g4`
**FEN:** `1rr3k1/5pp1/p1pbpq1p/P2p3b/Q2P2P1/2P2N1P/1P1N1P2/R3R1K1 b - - 0 18`

| section | before | after |
|---|---|---|
| On the card | Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 avoids it. Mistake: it cost 15% of your winning chances. | Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 moves your bishop on h5 away, where it had no defender. Mistake: it cost 15% of your winning chances. |
| What Bg6 does instead | 18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2 | it moves your bishop on h5 away, where it had no defender · 18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2 |
| The difference | After Rxb2, they win a pawn in the engine's line. Bg6 keeps it. | After Rxb2, they win a pawn in the engine's line. Bg6 keeps it: it moves your bishop on h5 away, where it had no defender. |
| Retry hint | *(not shown)* | Hint: it moves your bishop on h5 away, where it had no defender. The move starts on the lit square. |
| Retry answer | *(not shown)* | The move was Bg6. it moves your bishop on h5 away, where it had no defender |

**The judge (run 1, where the new text was Y):** "Y says Bg6 rescues the attacked, undefended bishop on h5 that gxh5 wins, where X only says Bg6 'avoids it'."

Figure 1: Round 3: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

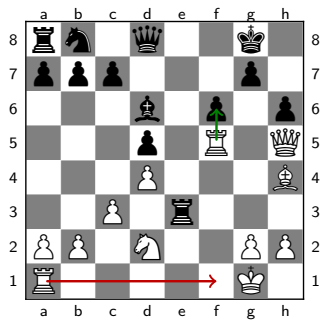


**Played:** 29. ♖d1, White to move; the app's verdict: Blunder.
**Engine:** +0.0 with the best move, −4.7 after yours.
**Best line:** 29. ♖e1 ♘e4 30. ♖b4 f5 31. ♖a4 ♖ee7
**UCI:** `f1d1`; best line `f1e1 d2e4 b5b4 f6f5 b4a4 e8e7`
**FEN:** `4r1k1/p2r2p1/5p1p/1R6/8/1P6/P2n1BPP/5RK1 w - - 1 29`

| section | before | after |
|---|---|---|
| On the card | Rd1 allows Nf3+, and it attacks the pawn on h2. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances. | Rd1 allows Nf3+, and if you take it with gxf3, Rxd1+ takes your rook on d1. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances. |
| Takeaway | Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it attacks the pawn on h2. | Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it if you take it with gxf3, Rxd1+ takes your rook on d1. |
| What Rd1 allowed: Nf3+ | it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3 | if you take it with gxf3, Rxd1+ takes your rook on d1; it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3 |

**The judge (run 1, where the new text was X):** "X names the real tactic, that Nf3+ comes with check and uncovers the d7 rook so Rxd1+ wins your rook on d1, while Y only says the knight attacks h2, which does not explain the lost material."

Figure 2: Round 4: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

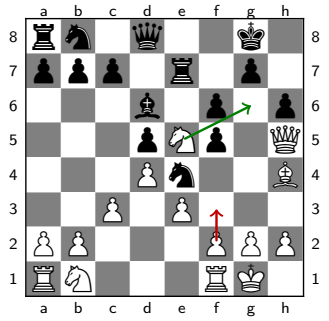


**Played:** 17. ♖af1, White to move; the app's verdict: Blunder.
**Engine:** +2.5 with the best move, 0.0 after yours.
**Best line:** 17. ♖xf6 ♕e8 18. ♕xd5+ ♔h8 19. ♖f7 ♘c6
**UCI:** `a1f1`; best line `f5f6 d8e8 h5d5 g8h8 f6f7 b8c6`
**FEN:** `rn1q2k1/ppp3p1/3b1p1p/3p1R1Q/3P3B/2P1r3/PP1N2PP/R5K1 w - - 0 17`

| section | before | after |
|---|---|---|
| What Rxf6 does instead | it attacks the bishop on d6 and the pawn on h6 · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7 | it attacks the bishop on d6 and the pawn on h6; if they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up; if they take it with Qxf6, Bxf6 wins their queen for your rook · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7 |
| Retry answer | The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6. | The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6. If they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up. If they take it with Qxf6, Bxf6 wins their queen for your rook. |

**The judge (run 1, where the new text was X):** "X shows why the rook sacrifice works (gxf6 Qg6+ Kf8 Qxh6+ then Qxe3 wins the rook back, and Qxf6 Bxf6 wins the queen), while Y leaves the player wondering why Rxf6 does not just lose a rook to gxf6."

Figure 3: Round 7: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

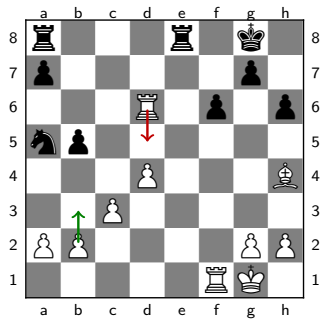


**Played:** 13. f3, White to move; the app's verdict: Inaccuracy.
**Engine:** +1.5 with the best move, +0.5 after yours.
**Best line:** 13. ♘g6 ♖e8 14. ♕xf5 c6 15. f3 ♘g5
**UCI:** `f2f3`; best line `e5g6 e7e8 h5f5 c7c6 f2f3 e4g5`
**FEN:** `rn1q2k1/ppp1r1p1/3b1p1p/3pNp1Q/3Pn2B/2P1P3/PP3PPP/RN3RK1 w - - 0 13`

| section | before | after |
|---|---|---|
| Takeaway | Look for the more active move first: Ng6 attacks the rook on e7. f3 can often still be played a move later. | Look for the more active move first: Ng6 attacks the rook on e7. f3 still comes, two moves later, in the engine's line. |
| The difference | You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7. | You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7. The order matters: f3 comes in the engine's line too, two moves later (Ng6 Re8 Qxf5 c6 f3). |

**The judge (run 1, where the new text was X):** "X says correctly that f3 still comes two moves later in the engine's line (Ng6 Re8 Qxf5 c6 f3), which shows the problem is move order; Y's generic 'can often still be played a move later' is vaguer and gets the count wrong."

Figure 4: Round 8: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

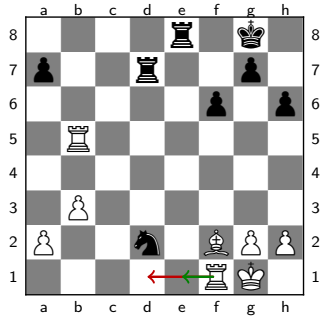


**Played:** 23. ♖d5, White to move; the app's verdict: Inaccuracy.
**Engine:** +2.1 with the best move, +1.0 after yours.
**Best line:** 23. b3 ♖e2 24. ♖a6 ♘b7 25. ♖e1 ♖c2
**UCI:** `d6d5`; best line `b2b3 e8e2 d6a6 a5b7 f1e1 e2c2`
**FEN:** `r3r1k1/p5p1/3R1p1p/np6/3P3B/2P5/PP4PP/5RK1 w - - 1 23`

| section | before | after |
|---|---|---|
| What b3 does instead | if they take it with Nxb3, axb3 wins their knight for your pawn · 23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8 | 23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8 |
| Retry answer | The move was b3 (the green arrow). If they take it with Nxb3, axb3 wins their knight for your pawn. | The move was b3 (the green arrow). |

**The judge (run 1, where the new text was X):** "Y adds the true safety check that b3 does not hang to the a5 knight, since Nxb3 axb3 wins the knight."

Figure 5: Round 9: the judge preferred the old text in both runs. The move played is the red arrow, the engine's best move the green one.

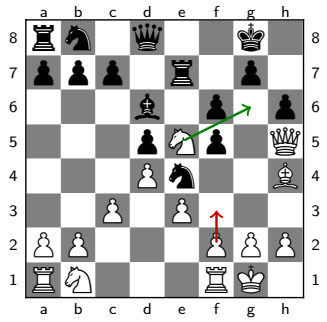


**Played:** 29. ♖d1, White to move; the app's verdict: Blunder.
**Engine:** +0.2 with the best move, −4.6 after yours.
**Best line:** 29. ♖e1 ♘e4 30. ♖a5 ♖ee7 31. b4 ♘xf2
**UCI:** `f1d1`; best line `f1e1 d2e4 b5a5 e8e7 b3b4 e4f2`
**FEN:** `4r1k1/p2r2p1/5p1p/1R6/8/1P6/P2n1BPP/5RK1 w - - 1 29`

| section | before | after |
|---|---|---|
| What Re1 does instead | it attacks the rook on e8 (undefended); the rook takes the open e-file · 29.Re1 Ne4 30.Ra5 Ree7 31.b4 Nxf2 32.Rxe7 Nh3+ | it attacks the rook on e8 (undefended); the rook takes the open e-file; if they take it with Rxe1+, Bxe1 takes back: an even trade · 29.Re1 Ne4 30.Ra5 Ree7 31.b4 Nxf2 32.Rxe7 Nh3+ |

**The judge (run 1, where the new text was X):** "X adds the true point that if Black trades with Rxe1+, Bxe1 recaptures for an even trade, which shows why Re1 is safe."

Figure 6: Round 10: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

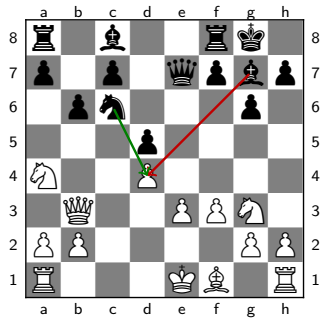


**Played:** 13. f3, White to move; the app's verdict: Mistake.
**Engine:** +1.6 with the best move, +0.3 after yours.
**Best line:** 13. ♘g6 ♖e8 14. ♕xf5 c6 15. f3 ♘g5
**UCI:** `f2f3`; best line `e5g6 e7e8 h5f5 c7c6 f2f3 e4g5`
**FEN:** `rn1q2k1/ppp1r1p1/3b1p1p/3pNp1Q/3Pn2B/2P1P3/PP3PPP/RN3RK1 w - - 0 13`

| section | before | after |
|---|---|---|
| What Ng6 does instead | it attacks the rook on e7 · 13.Ng6 Re8 14.Qxf5 c6 15.f3 Ng5 16.Bf2 a5 | it attacks the rook on e7; if they move the rook (Rf7), Qxf5, and you have won a pawn · 13.Ng6 Re8 14.Qxf5 c6 15.f3 Ng5 16.Bf2 a5 |

**The judge (run 1, where the new text was X):** "X adds the concrete point that once the attacked rook moves (Rf7), Qxf5 wins the f5 pawn, tying Ng6's threat to the material it gains."

Figure 7: Round 12: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.



**Played:** 14. . . ♗xd4, Black to move; the app's verdict: Inaccuracy.
**Engine:** +5.0 with the best move, +3.5 after yours.
**Best line:** 14. . . ♘xd4 15. ♕d3 ♗f5 16. ♕d2 ♘c2+ 17. ♔f2
**UCI:** `g7d4`; best line `c6d4 b3d3 c8f5 d3d2 d4c2 e1f2`
**FEN:** `r1b2rk1/p1p1qpbp/1pn3p1/3p4/N2P4/1Q2PPN1/PP4PP/R3KB1R b KQ - 1 14`

| section | before | after |
|---|---|---|
| What Nxd4 does instead | it attacks the queen on b3 and the pawn on f3; it wins the pawn outright; if they move the queen (Qxd5), Qxe3+ Be2 Nc2+ Kf1 Bd4 Ne4 Bf5 Nac3 Nxa1, and you have won about 6 pawns of material · 14...Nxd4 15.Qd3 Bf5 16.Qd2 Nc2+ 17.Kf2 Nxa1 18.Nxf5 | it attacks the queen on b3 and the pawn on f3; it wins the pawn outright; if they move the queen (Qxd5), you have won about 6 pawns of material by move 19 · 14...Nxd4 15.Qd3 Bf5 16.Qd2 Nc2+ 17.Kf2 Nxa1 18.Nxf5 |

**The judge (run 1, where the new text was X):** "Y shows the forcing sequence after Qxd5 (Qxe3+, Nc2+ and finally Nxa1) that wins the rook, while X only says it happens 'by move 19'."

Figure 8: Round 13: the judge preferred the old text in both runs. The move played is the red arrow, the engine's best move the green one.

## 6 Results

From the baseline to the kept variant (v8), the last one graded on the rounds' own documents, on the held-out games:

- faithful: 90.1 [84.2, 95.9]% to 100.0 [100.0, 100.0]%, a paired change of +9.9 [4.0, 15.9] points;
- concrete: 75.0 [65.3, 83.6]% to 82.9 [72.0, 91.4]%, a paired change of +7.9 [3.9, 12.1] points;
- named_deep: 94.1 [89.7, 97.9]% throughout, as expected of a property of the evidence;
- faithfulAll: 89.5 [83.3, 95.2]% to 99.3 [97.8, 100.0]%.

On train the same changes read 87.6% to 99.4% (faithful, +11.9 [7.9, 16.4]) and 78.0% to 84.7% (concrete, +6.8 [3.2, 10.6]). The kept variant changes the visible texts of 70 test and 87 train moments. Every kept round the judge read (all but rounds 1, 2, 11 and 14, which fix errors) was preferred on both halves; round 5 was not, and rounds 9 and 13 were rejected nearly unanimously.

Round 10's sentence makes no claim the grades check. Its round was graded on the 329 moments of documents rebuilt for it, against its own base on the same documents, and the two agree on every grade (Table 4); its texts differ from the base's on 6 test and 10 train moments. So the rates above stop at v8.

Round 11 was graded the same way, on round 10's rebuilt documents against round 10's texts. It changes the texts of 35 test and 42 train moments; no grade falls, and the moments graded concrete rise from 126 to 127 on test and from 148 to 149 on train (Table 4). Of the 59 defects `defects.mjs` counts in round 10's texts, none is left (Table 5).

Round 12 was graded on documents rebuilt for it at round 11's code, with fresh capture searches and its own defense searches, against round 11's texts on the same documents. It changes the texts

of 17 test and 14 train moments, each of them graded faithful, and no grade moves (Table 4). The judge preferred the new texts 19–9 on train and 22–12 on test, with no drift toward length in the padding control dealt into the same batches. Its reasons against the new texts were mostly their length: pooling both halves, the new text won 25–5 where the sentence shows 4 moves or fewer, and 16–16 where it shows 5 or more. A round that shortens the line has seen that tally, so it fixes its cut on train.

Round 13 did, and was graded on the documents the reproduction rebuilt at the app then pinned (Section 7), against that app's texts on them, after the grader learned its two new forms and graded the base exactly as before. It changes the texts of 14 test and 8 train moments, each graded faithful, and no grade moves (Table 4). It barely shortened them: by a median of 14 characters, and 7 of the 22 grew, where the pointer is longer than the short line it replaced. The judge preferred the old texts, 0–16 on train and 4–24 on test: 1–21 where the sentence points to the engine's line, and 3–19 where it leaves out a long line. Its reasons name the moves: they show how the material falls, and a move number gives the player nothing to follow or check. The pointer was also worse than it looked: in 4 of the 11 sentences that point to the engine's line, the gain comes after the moves the lesson prints beside it, and a searched defense's line is not on the card at all. Round 12's long lines had split the judge evenly; set against the same sentence without its moves, it wanted the moves.

Round 14 was graded on the documents round 13 was, against that app's texts on them, after the grader learned to read the pieces behind each amount and the round's new phrases. Every amount the base says is the line's net, which is what faithful checks, so its misnamed pieces passed; no grade moves with the round (Table 4). One of the grader's fixes came after it first graded the round: it read "a rook for a pawn" as "a rook" where an amount ends a sentence, and failed those amounts. With the longer phrases tried first, it grades the base exactly as before, as it did after each of its other changes. What the round fixes is counted instead (Table 6): in 51 of the base's moments an amount names pieces other than the ones that changed hands, and in 10 a piece said to fall is given back later in the line. In round 14's texts neither is left and none of round 11's defects appears, and the texts change on those 58 moments (32 test, 26 train) and on no other.

Round 15 was graded on the same documents, against the texts of the app then pinned, which carry round 14, after the grader learned the round's new clauses, each checked on the board apart from the app's code, and graded the base exactly as before. It changes the texts of 4 test and 12 train moments, each graded faithful; no grade falls, and the moments graded concrete rise from 127 to 128 on test and from 149 to 152 on train (Table 4). The judge preferred the new texts 22–1 on train, with 1 same, and 8–0 on test. One moment did not win both runs: there the reply guards a pawn the better move would have attacked, but the played move attacks that pawn too and the engine answers the better move with the same reply, so the point does not tell the two moves apart.

Round 16 was graded on the same documents, against the texts of the app then pinned, which carry round 15, and on a second set built for it. The 60 games hold few endgame moments, so the round adds an endgame set: 160 more of the player's games from the routine's archive, outside the 60, that reach the endgame by the app's divider, drawn with a fixed seed and split by game, 80 train and 80 test, before any lesson was read. Its key moments in the endgame, 219 train and 184 test, were built at that app on the lab's evidence with the routine's searches. The grader learned the round's new sentences first, each checked on the board apart from the app's code, and graded the base exactly as before in both sets. The round changes the texts of 19 train and 3 test moments of the endgame set and of 2 train and 8 test moments of the 60 games, each graded faithful; no moment loses a grade, and three become concrete (Table 4). Judging both sets together, each moment in its own set's split, the judge preferred the new texts 37–2 on train, with 3 same, and 18–3 on test, with 1

same. Two moments lost both runs: in one the king's walk the lesson names comes in the better line too, so it does not tell the moves apart; in the other the "walk" is a king stepping out of checks and taking back a queen.

Round 17 was graded on the same documents in both sets, against the texts of the app then pinned, which carry round 16, after the grader learned to check its sentence against the stored scores apart from the app's code and graded the base exactly as before. It changes the texts of 188 train and 159 test moments across the two sets, and no grade moves (Table 4): every one of its sentences reads true. That is more moments than a judge can read, so the judges read a sample drawn from the changed moments' ids alone, before any was read: in each set and split, 20 taken by a seeded shuffle (seed 20261015), 80 in all. On round 17's first wording the judge preferred the new texts in every comparison it decided, 72 on train, with 8 same, and 74 on test, with 6 same. Round 17b judged again the 49 sampled moments whose wording it changes, and on the final texts the judge preferred the new ones 73–0 on train, with 7 same, and 76–0 on test, with 4 same. Every same-answer is on the form that says both moves leave the mover ahead or behind, and all came from the second run's judges, who found the sentence true but no help in seeing why the better move is better. The player's five third-set moments the draw left out were judged in the same batches and are not counted: the new text won 10 of their 10 comparisons.

| variant | split | moments | faithful | concrete | named_deep | faithfulAll | texts changed |
|---|---|---|---|---|---|---|---|
| v10 | test | 152 | 152 | 126 | 143 | 151 | 6 |
| v10 | train | 177 | 176 | 148 | 163 | 176 | 10 |
| v11 | test | 152 | 152 | 126 / 127 | 143 | 151 | 35 |
| v11 | train | 177 | 176 | 148 / 149 | 163 | 176 | 42 |
| v12 | test | 152 | 152 | 127 | 143 | 151 | 17 |
| v12 | train | 177 | 176 | 149 | 163 | 176 | 14 |
| v13 | test | 153 | 153 | 127 | 144 | 152 | 14 |
| v13 | train | 180 | 179 | 149 | 166 | 179 | 8 |
| v14 | test | 153 | 153 | 127 | 144 | 152 | 32 |
| v14 | train | 180 | 179 | 149 | 166 | 179 | 26 |
| v15 | test | 153 | 153 | 127 / 128 | 144 | 152 | 4 |
| v15 | train | 180 | 179 | 149 / 152 | 166 | 179 | 12 |
| v16 | test | 153 | 153 | 128 / 129 | 144 | 152 | 8 |
| v16 | train | 180 | 179 | 152 | 166 | 179 | 2 |
| v17 | test | 153 | 153 | 129 | 144 | 152 | 65 |
| v17 | train | 180 | 179 | 152 | 166 | 179 | 72 |
| v17b | test | 153 | 153 | 129 | 144 | 152 | 65 |
| v17b | train | 180 | 179 | 152 | 166 | 179 | 72 |
| v16, endgame set | test | 184 | 181 | 162 | 158 | 180 | 3 |
| v16, endgame set | train | 219 | 213 | 170 / 172 | 204 | 213 | 19 |
| v17, endgame set | test | 184 | 181 | 162 | 158 | 180 | 94 |
| v17, endgame set | train | 219 | 213 | 172 | 204 | 213 | 116 |
| v17b, endgame set | test | 184 | 181 | 162 | 158 | 180 | 94 |
| v17b, endgame set | train | 219 | 213 | 172 | 204 | 213 | 116 |

Table 4: A round graded on rebuilt documents, beside its base on the same documents: moments graded 1, with one number where the round and its base agree. The rows marked endgame set are rounds 16 and 17 on the second set; v17 is round 17's first wording and v17b its last.

| defect | v10 | v11 |
|---|---|---|
| a later sentence of the retry reveal in lower case | 29 | 0 |
| a recapture called a win “outright” | 15 | 0 |
| a capture or a check called “a quiet improvement” | 6 | 0 |
| “too short or unsettled” beside a capture the card calls outright | 3 | 0 |
| a lost piece worth more than the amount named | 6 | 0 |

Table 5: The text defects round 11 fixes, counted in each variant's texts on every moment of round 10's rebuilt documents (`defects.mjs`).

| defect | base | v14 |
|---|---|---|
| an amount that names pieces other than the ones that changed hands | 51 | 0 |
| a piece said to fall that the line gives back | 10 | 0 |
| the five of Table 5 | 0 | 0 |

Table 6: The text defects round 14 fixes, counted in its base's texts and round 14's on every moment of round 13's documents (`defects.mjs`), with round 11's five beside them.

# 7 Reproduction in the lab

**Recomputed.** The lab recomputes every variant's aggregates from the per-moment grades the records keep (`summarize.mjs`): 27 of the 27 lines of the published aggregates match exactly. Every judge round's tally is recomputed from the judge's answers and the order keys (`judge/pairwise.mjs`), and every decision follows from the rule as recorded. One recorded reason misstates its counts: round 5's reads “train 4 new vs 6 old (test 10 vs 2): the picture hurt mates of 8 to 13 moves”, while the recorded answers give 6–6 on train and 14–2 on test. The new counts it gives are those of the moments round 6 later kept; the decision is the same, since a tie on train is not a majority.

**Rerun.** The lab rebuilt the engine evidence for the same games on the app as it now runs (`d6e155b`, evidence review-evidence/6), redid the capture searches on it (88 kept) and the defense searches (51 kept), and graded the app's texts. The app at that commit carries the kept rounds through round 16; round 17 is not in it yet. Its texts equal the recorded texts of v17b, the last kept round, on 196 of the 333 moments of round 17's documents. The rebuilt evidence is not the hill-climb's: the review pipeline changed after the rounds, so engine lines differ in detail and some key moments move (on test, 146 of 153 moments are the same moments; on train 170 of 180). Every rate lies within the recorded v8's 95% interval (Table 7). The records are read at `3429b1e`.

| split | grade | recorded baseline | recorded v8 | pinned app, rerun |
|---|---|---|---|---|
| test | faithful | 90.1 [84.2, 95.9] | 100.0 [100.0, 100.0] | 100.0 [100.0, 100.0] |
| test | concrete | 75.0 [65.3, 83.6] | 82.9 [72.0, 91.4] | 84.3 [76.7, 91.0] |
| test | named_deep | 94.1 [89.7, 97.9] | 94.1 [89.7, 97.9] | 94.1 [89.7, 97.9] |
| test | faithfulAll | 89.5 [83.3, 95.2] | 99.3 [97.8, 100.0] | 99.3 [97.8, 100.0] |
| train | faithful | 87.6 [83.0, 91.7] | 99.4 [98.1, 100.0] | 99.4 [98.2, 100.0] |
| train | concrete | 78.0 [70.7, 84.2] | 84.7 [78.9, 89.8] | 84.4 [78.2, 89.9] |
| train | named_deep | 92.1 [86.8, 97.1] | 92.1 [86.8, 97.1] | 92.2 [87.0, 97.1] |
| train | faithfulAll | 87.6 [82.9, 91.8] | 99.4 [98.2, 100.0] | 99.4 [98.2, 100.0] |

Table 7: The recorded baseline and kept variant beside the pinned app's texts on rebuilt evidence.

# 8 Limitations

- The grades check what can be checked: a true and concrete text can still fail to teach. That question goes to the player on the Calibration page, and to the judge, whose agreement with the player is itself only partly measured.
- The length check shows that the judge does not reward words that say nothing. It does not show that the judge can tell a point that helps from a true point that does not, at the same length; a decoy built from true but beside-the-point facts about the position would test that.
- The judge is a model reading on the player's behalf. It is a good finder of what is missing and a poorer judge of how much it matters, which is why the rounds use its pairwise preferences and its notes, not its ratings.
- Round 12 answers one defense, the model's likeliest; when the defender's second choice is nearly as likely, the lesson leaves it unanswered.
- Round 16 names a king's walk in one line without asking whether the other line makes it too; where both do, the point does not tell the moves apart. What separates such moves is often a king cut off along a file, which no reading names yet.
- Round 17's judges read a sample of the moments it changed, 80 of 347, so its verdict is the sample's. Its sentence that both moves leave the mover ahead or behind is the one the judges did not always prefer.
- The app's folded note on how a grade was made rounds a score's tenth differently from round 17's sentence, so on some scores the card gives a move two numbers a tenth apart.
- 60 games of one player. The split keeps the claims honest about that player, not about players in general.